JNTUH USED 23-05-18 AM Set 2

Code No: 117AN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, April/May - 2018
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) What is an intelligent system? [2]
b) What do you mean by an exhaustive search? Give examples of exhaustive search. [3]
c) What are the basic components in a frame? [2]
d) Distinguish between forward and backward chaining. [3]
e) What is an expert system shell? [2]
f) What is a Bayesian network? [3]
g) Distinguish between classification and clustering. [2]
h) Distinguish between inductive and deductive learning. [3]
i) Distinguish between syntax and semantics in natural language processing. [2]
j) What do you mean by natural language understanding? [3]

PART-B

(50 Marks)

2.a) Define the following terms:
i) Problem space ii) Problem solving iii) State space iv) State space search
b) What are the characteristics of a heuristic search? Compare brute force and heuristic search. [5+5]

OR

3. Demonstrate A* algorithm with an example and explain admissibility of A* [10]

4.a) What are the advantages and problems of a frame based knowledge representation? Give examples of knowledge hard to represent in a frame.
b) Give a simple resolution algorithm for propositional logic. [5+5]

OR

5.a) Explain the basic inference mechanism in semantic nets and frames.
b) Prove the inconsistency of $(a \vee b) \& (e \vee f) \& (\sim a \vee b) \& \sim b$ using semantic tableaux method. [5+5]

6.a) What are the basic components of an expert system? Explain.
b) What is the purpose of a Truth Maintenance System in expert systems? [5+5]

OR

7.a) Explain the general approach for developing an expert system.
b) Give an overview of rule-based system using certainty factor. [5+5]

8. Explain maximum margin hyperplane and linear Support Vector Machine. [10]

OR

9. What do you mean by linear separability? Give an overview of perceptron and perceptron training algorithm. [10]

10. What is a grammar? Give an overview of various methods for specifying grammar in natural language processing. [10]

OR

11.a) Give an overview of syntax representation and parsing methods in natural language processing.

b) What is a semantic web? What are the challenges automated reasoning systems will have to deal with respect to semantic web? [5+5]

--ooOoo--

JNTUH USED 23-05-18AM Set 2